

# Gradient-boosted trees

HistGradientBoosting on the tabular features: accuracy and feature-importance reference.

Gradient-boosted trees are a strong tabular reference but far too large for the microcontroller (about 8 bytes per node).

## Architecture

- scikit-learn HistGradientBoostingRegressor: squared error, learning rate 0.05, up to 500 trees of depth 6 (31 leaves), min 100 samples per leaf, L2 1.0, early stopping on the validation split.
- 500 trees, 30,500 nodes: about 238 KB at 8 B per node, far over the 32 KB budget, and about 3,000 comparisons per prediction.
- Residual over the RLS forecast like the neural models.

## Inputs and feature engineering (gbdt)

- **Tabular input:** the engineered features of the newest beat (see 01\_feature\_engineering.pdf), z-scored per feature with training statistics.
- **Target:**  $dRR[n+1] = RR[n+1] - RR[n]$ . Residual models predict the z-scored correction over the RLS forecast ( $dRR - rls\_pred$ ); the output is de-normalised and added back to  $rls\_pred$ .

### Tabular features (19):

| feature     | definition                                                                                  |
|-------------|---------------------------------------------------------------------------------------------|
| rr_dev      | newest RR minus the window mean                                                             |
| rr_mean     | mean RR of the 30-beat window                                                               |
| drr_lag0..7 | drr_lag0..7: the 8 most recent dRR (lag0 = newest)                                          |
| rmssd       | root mean square of the 29 successive differences                                           |
| rs          | rescaled range: (max - min of the cumulative sum of RR - mean) / std                        |
| ccm         | complex correlation measure: Poincare-triangle area / (pi SD1 SD2 (W-3)), clipped to [0, 1] |
| guzik       | Guzik asymmetry: sum of positive dRR / sum of  dRR                                          |
| rls_pred    | RLS AR(8) forecast of the next dRR (ms), clamped to +300                                    |
| rls_err     | last RLS innovation: true dRR minus the previous forecast (ms)                              |
| bp          | RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1                    |
| bp_prev     | previous band-pass output (gives the phase with bp)                                         |
| run_len     | signed length of the current run of same-sign dRR (+ = decelerating)                        |

## Variants trained

| run      | change vs parent                      | feature set (n)  | params | MACs/beat | training               | epochs |
|----------|---------------------------------------|------------------|--------|-----------|------------------------|--------|
| gbdt     | 19 tabular features                   | tab (19)         | -      | -         | from scratch, lr 0.001 | -      |
| gbdt_all | every engine feature (29 at the time) | 29 features (29) | -      | -         | from scratch, lr 0.001 | -      |

## Results

| model    | val open loop | closed-loop model | 10%   | 30%   | 50%   | 80%   | RSA invented at 80% | test whole-series 80% | val source |
|----------|---------------|-------------------|-------|-------|-------|-------|---------------------|-----------------------|------------|
| gbdt     | 34.19         | float             | 25.53 | 29.94 | 33.71 | 43.16 | -                   | 45.85                 | round 1    |
| gbdt_all | -             | float             | -     | -     | -     | -     | -                   | 45.51                 | -          |

RMSE in ms. **val open loop:** one-step-ahead on the true history, 16 validation subjects. **10-80%:** closed loop (pacemaker use) on validation: that share of beats dropped at random from beat 0, each filled by the model and fed back; RMSE on model-filled beats, first 5000 beats per subject, mean of the seeds of that round (3 seeds from round 2 on). int8 is used where it exists. **RSA invented:** share of subject-runs whose filled series gains an RSA peak the original lacks. **test whole-series 80%:** diagnostic on the 10 test subjects (whole series, seed 7, results/test/ folder); it was not used for any selection.

## Correlation between the original and the filled series (validation and test subjects)

| model    | corr RR validation | corr RR test | corr dRR validation | corr dRR test | RMSE filled, validation (ms) | RMSE filled, test (ms) |
|----------|--------------------|--------------|---------------------|---------------|------------------------------|------------------------|
| gbdt     | 0.9312             | 0.9359       | 0.248               | 0.244         | 51.99                        | 45.85                  |
| gbdt_all | 0.9321             | 0.9365       | 0.257               | 0.250         | 51.62                        | 45.51                  |

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = `np.corrcoef(original_serie, modified_serie)[0, 1]`; **corr dRR** = the same function on the beat-to-beat changes, `np.corrcoef(np.diff(original), np.diff(filled))[0, 1]`, which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled**: RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

## Verdict

- Not deployable (size). Accuracy reference only: 34.2 ms on validation, between the MLP and the TCNs.
- gbd\_t\_all (34.15 ms) shows that the extra features add nothing; its permutation importance ranks `rls_pred` first by far, then `rr_dev`, `drr_lag0`, `guzik` and `bp`; the 8 RLS weights contribute nothing.

## Abbreviations and acronyms used in this document

| term               | meaning                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AR</b>          | autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values                                                                              |
| <b>B</b>           | batch dimension (number of windows processed together) in the layer output shapes                                                                                         |
| <b>bp</b>          | feature: RSA band-pass output: RBJ biquad on raw RR, centre 0.30 cycles/beat, Q 1                                                                                         |
| <b>bp_prev</b>     | feature: previous band-pass output (gives the phase with bp)                                                                                                              |
| <b>Burg-AR</b>     | AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)                                                                                  |
| <b>ccm</b>         | complex correlation measure: non-linearity index of the Poincare plot of consecutive RR intervals                                                                         |
| <b>dRR</b>         | beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted                                                                         |
| <b>drr_lag0..7</b> | feature: drr_lag0..7: the 8 most recent dRR (lag0 = newest)                                                                                                               |
| <b>gbd</b>         | gradient-boosted decision trees (run name)                                                                                                                                |
| <b>guzik</b>       | feature: Guzik asymmetry: sum of positive dRR / sum of  dRR                                                                                                               |
| <b>HRV</b>         | heart rate variability                                                                                                                                                    |
| <b>int8</b>        | 8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run                                      |
| <b>KB</b>          | kilobyte (1,024 bytes)                                                                                                                                                    |
| <b>L2</b>          | L2 regularisation: a penalty on the sum of squared model weights                                                                                                          |
| <b>lr</b>          | learning rate (step size of the optimiser)                                                                                                                                |
| <b>MAC</b>         | multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat |
| <b>MACs</b>        | multiply-accumulate operations (see MAC)                                                                                                                                  |
| <b>MLP</b>         | multilayer perceptron (a fully connected neural network)                                                                                                                  |
| <b>ms</b>          | milliseconds                                                                                                                                                              |
| <b>np.corrcoef</b> | NumPy's Pearson correlation-coefficient function                                                                                                                          |
| <b>np.diff</b>     | NumPy's successive-difference function                                                                                                                                    |
| <b>Q</b>           | quality factor of a band-pass filter: centre frequency / bandwidth                                                                                                        |
| <b>RBJ</b>         | Robert Bristow-Johnson's standard biquad (second-order) filter formulas                                                                                                   |
| <b>RLS</b>         | recursive least squares: an adaptive filter that updates its coefficients at every beat                                                                                   |
| <b>rls_err</b>     | feature: last RLS innovation: true dRR minus the previous forecast (ms)                                                                                                   |
| <b>rls_pred</b>    | feature: RLS AR(8) forecast of the next dRR (ms), clamped to +/-300                                                                                                       |
| <b>RMSE</b>        | root mean square error (ms): the main accuracy measure, lower is better                                                                                                   |
| <b>rmssd</b>       | root mean square of successive differences of the RR intervals in the window                                                                                              |
| <b>RR</b>          | RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms                                                                                          |
| <b>rr_dev</b>      | feature: newest RR minus the window mean                                                                                                                                  |
| <b>rr_mean</b>     | feature: mean RR of the 30-beat window                                                                                                                                    |
| <b>rs</b>          | rescaled range: (max - min of the cumulative deviations from the mean) / standard deviation                                                                               |
| <b>RSA</b>         | respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing                                                                                       |
| <b>run_len</b>     | feature: signed length of the current run of same-sign dRR (+ = decelerating)                                                                                             |
| <b>SD1</b>         | Poincare-plot standard deviation across the identity line (short-term variability)                                                                                        |
| <b>SD2</b>         | Poincare-plot standard deviation along the identity line (long-term variability)                                                                                          |
| <b>tab</b>         | feature-set / model-kind name: tabular model (engineered features only)                                                                                                   |
| <b>TCN</b>         | temporal convolutional network (causal, dilated 1-D convolutions)                                                                                                         |
| <b>val</b>         | validation: the 16 validation subjects, used for model selection                                                                                                          |
| <b>W</b>           | window length: the 30 most recent RR intervals                                                                                                                            |
| <b>z</b>           | z-score: (value - mean) / standard deviation, with statistics from the training subjects                                                                                  |